

Emotion Synthesizer: An Immersive XR Art Installation Integrating Real and Virtual Emotions

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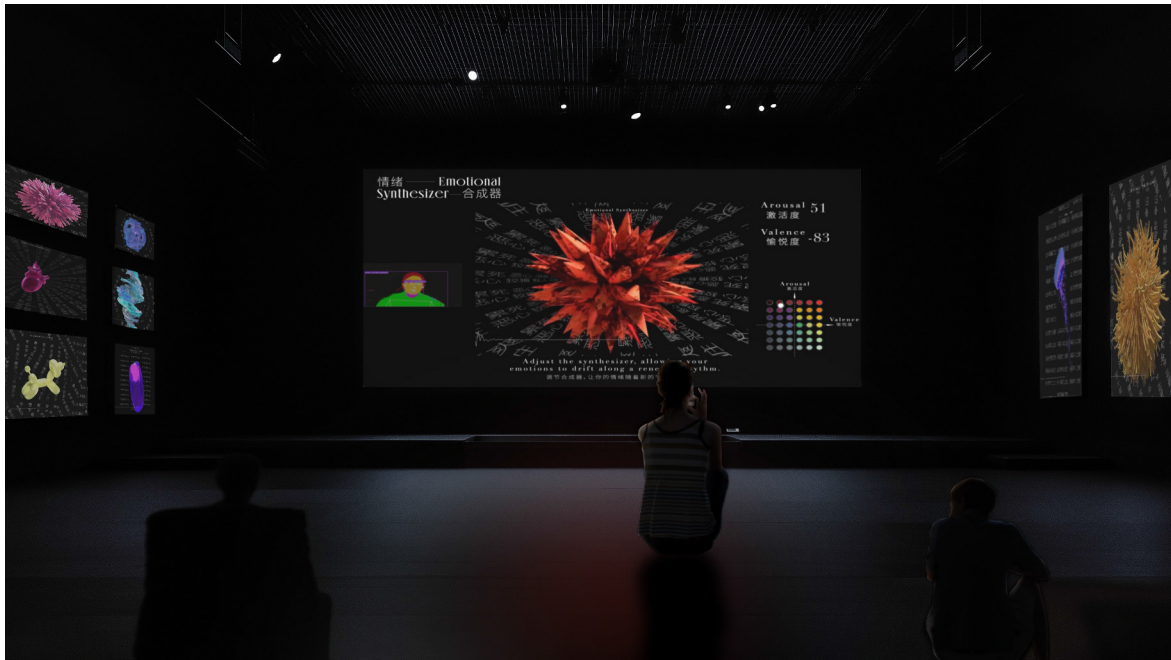
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Figure 1: Exhibition View of Emotion Synthesizer

ABSTRACT

As technology increasingly mediates our emotional experiences, the boundaries between authentic feelings and digitally constructed states become increasingly fluid. Emotion Synthesizer is an immersive art installation that investigates this evolving emotional landscape through the lens of extended reality. The installation combines XR technology, interactive visualization, and AI-generated content to create an immersive environment where viewers can observe and interact with their emotional states. Through real-time interaction and visualization, the installation establishes a unique dialogue between real and virtual emotional experiences. This work raises critical questions about emotional authenticity, self-perception, and the impact of digital information on emotional experiences in the contemporary technological era.

Index Terms: Digital reality, AI-generated art, XR art installation, Emotional authenticity, Emotion visualization.

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1 DESCRIPTION OF ARTWORK

In an age of rapid technological advancement and overwhelming information flows, emotional states are increasingly shaped and manipulated by external data streams, as personal emotions become ever more data-driven and constructed. Emotion Synthesizer is an immersive, interactive visualization art installation exploring the impact of information overload and intelligent data manipulation on human emotions. This work uses emotion models to help viewers recognize and understand their emotions in an immersive environment. Through an emotion agent, viewers can actively control virtual emotional scenes, blending real and virtual emotions through a combination of visuals and music, and drawing them into a space where real and virtual emotions merge.

Emotion Synthesizer delivers the XR experience through an immersive media environment. An LED screen fills the audience's view, while a physical synthesizer knob serves as the primary control, triggering synchronized visual and audio feedback. Multi-channel projection and surround sound create an enveloping audiovisual landscape, enabling deep engagement and real-time emotional adjustments. The theoretical foundation is based on Plutchik's Wheel of Emotions and Thayer's Valence-Arousal (VA) Model, which encodes emotional states through color, shape, and brightness, creating an emotional coordinate system. AI-generated 3D models serve as an emotion library mapped to this coordinate system, while pre-designed AI-generated musical fragments correspond to various emotional states, with parameters controlling au-

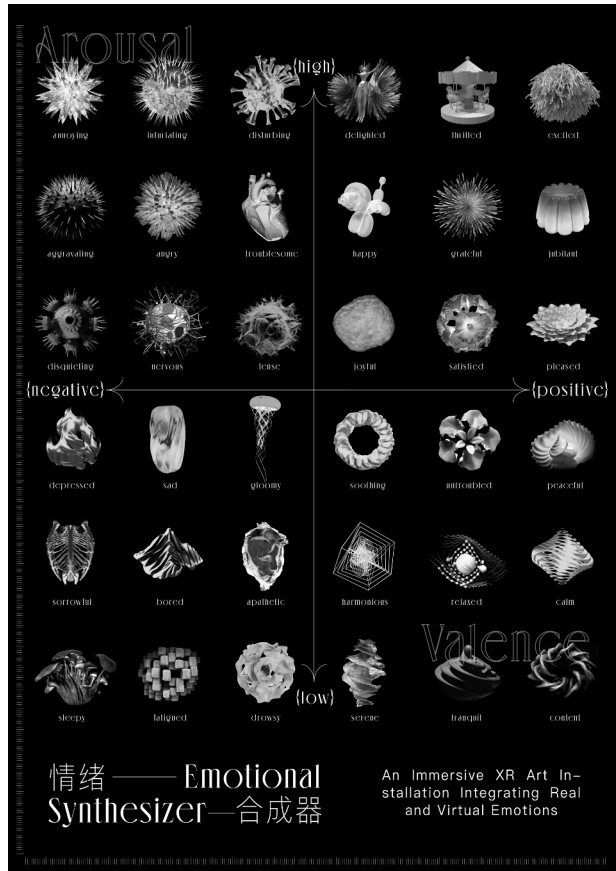


Figure 2: Virtual Emotional Model Representing Valence and Arousal

dio tempo, timbre, and other qualities to ensure alignment with visual transformations. The system synthesizes virtual and real emotional models by adjusting the activation and intensity values with the knob, constructing a complex emotional interaction space.

As viewers enter the exhibition space, a camera recognizes facial expressions and displays an emotional model reflecting each viewer's emotional activation and intensity. By externalizing their current emotional state as a visual model, viewers can observe and reflect on their emotions more objectively, enabling active adjustment or transformation. This visualization promotes emotional self-awareness, which is essential for effective emotional regulation. Viewers interact with their virtual emotional model using the synthesizer knob, which synchronously changes the color, shape, texture, and music of the emotional model. This multi-sensory stimulation guides viewers' emotional trajectory, fostering a sense of "control" over their emotions. This preserved three-dimensional visualization serves as an "information memory," allowing viewers to observe how their emotions evolve within an informational context, prompting contemplation on the dynamic interplay between self, emotion, and information.

This process helps individuals recognize that emotions can be adjusted and embraced. Viewers observe how their emotions fluctuate in response to environmental changes, prompting reflection on the authenticity of self and emotion: Are emotions experienced in a simulated virtual environment truly "real"? Does the ability to control emotions change your perception of self? If emotions are so easily modifiable, does this mean they are largely shaped by external information? Does this virtual information become part of

the "self," or is it merely a temporary state shaped by the external environment?

This real-time emotional feedback mechanism creates a new form of interaction at the boundary between virtual and real emotional experiences. It integrates viewers' emotional perceptions, feedback, and interactions with the environment. This approach transcends the boundaries between physical and digital realms, allowing users to experience both a tangible physical space and a digitally expressed emotional space, providing an immersive experience beyond the physical world. Through the XR-mediated emotional experience, XR technology becomes a powerful interface for perceiving and expressing emotion, prompting viewers to deeply reflect on the relationships between information, emotion, and self, as well as on their sense of emotional authenticity.

2 ARTIST INFORMATION

Rui Zhang is a postgraduate student in the Department of Science Popularization Information Design at the Academy of Arts and Design, Tsinghua University. Her main research areas include art therapy, artificial intelligence design, and immersive experience design. She is committed to combining artificial intelligence with artistic creation, changing the emotions and perceptions of the audience through immersive experiences, and promoting the in - depth integration of art and technology.

Yan Dong is a transdisciplinary artist and Ph.D. candidate at the Academy of Arts and Design, Tsinghua University, focusing on mixed reality experience, AI-assisted design, and digital craft. Prior to her research, she worked as an interactive game designer at ByteDance. Her research has been published in academic journals including Leonardo, Decoration, and CHI, among others. Additionally, she has also designed cover artworks for high-impact scientific journals such as Nature Medicine and Nano Energy. Her artistic installations have been exhibited at the China National Communication Center for Science and Technology and other venues.

Lanxi Xiao is an artist, designer, and TEDx speaker based in China. She is pursuing a PhD in Art and Technology at the Academy of Arts and Design, Tsinghua University. Her research focuses on AI-driven interactive installations and immersive XR art experiences, integrating AI, LLM and digital human technologies to explore human-AI collaboration. Lanxi's works have been exhibited at the 6th Art and Science International Exhibition, the National Exhibition and Convention Center, the National Indoor Stadium and Beijing Science Week. Lanxi is a reviewer for the 2nd eXplainable AI for the Arts workshop at ACM C&C 2024.

Peiyan Zou is a junior majoring in Art and Technology at the Academy of Arts and Design, Tsinghua University. His areas of expertise include interactive design, user interface design, animation production, and gamification design. Peiyan interned at ByteDance in design, participated in game design contests, and gained experience in team-based and VR game development, demonstrating his dedication to integrating technology with art by applying technical skills to creative expression and transforming artistic design into widely cherished applications.

Qiong Wu is a tenured professor at the Academy of Arts and Design, Tsinghua University, serving as Deputy Director of Tsinghua-Qingshang Institute and Director of Data Intelligence Design Institute. Her research focuses on data-driven innovation design and mixed reality user experience design, with over 40 papers published in core journals and international conferences. Her artistic works have been exhibited at prestigious venues including Ars Electronica Festival and International Art and Science Exhibition. She has led more than 40 innovative design projects with institutions.

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